

Creation Date 23-Jan-2009

Revision Date 03-Jan-2021

Revision Number 6

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                           |                                   |
|---------------------------|-----------------------------------|
| Product Description:      | <b>DIMETHYL SULFOXIDE</b>         |
| Cat No. :                 | <b>SP/2696/27</b>                 |
| Synonyms                  | Dimethyl sulfoxide; DMSO          |
| CAS-No                    | 67-68-5                           |
| EC-No.                    | 200-664-3                         |
| Molecular Formula         | C <sub>2</sub> H <sub>6</sub> O S |
| Reach Registration Number | -                                 |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

|                |                                                                                                                                                                                                                                                          |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Company        | <b>EU entity/business name</b><br>Acros Organics BVBA<br>Janssen Pharmaceuticaaan 3a<br>2440 Geel, Belgium<br><br><b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road, Loughborough,<br>Leicestershire LE11 5RG, United Kingdom |
| E-mail address | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                           |

### 1.4. Emergency telephone number

Tel: 01509 231166  
 Chemtrec US: (800) 424-9300  
 Chemtrec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

CLP Classification - Regulation (EC) No 1272/2008

Physical hazards

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

Based on available data, the classification criteria are not met

## **Health hazards**

Based on available data, the classification criteria are not met

## **Environmental hazards**

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## **2.2. Label elements**

Combustible liquid

## **2.3. Other hazards**

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

DMSO readily penetrates skin and may carry other dissolved chemicals into the body

Toxic to terrestrial vertebrates

## **SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS**

### **3.1. Substances**

| Component          | CAS-No  | EC-No.            | Weight % | CLP Classification - Regulation (EC) No 1272/2008 |
|--------------------|---------|-------------------|----------|---------------------------------------------------|
| Dimethyl sulfoxide | 67-68-5 | EEC No. 200-664-3 | >95      | -                                                 |

Reach Registration Number

-

Full text of Hazard Statements: see section 16

## **SECTION 4: FIRST AID MEASURES**

### **4.1. Description of first aid measures**

#### **General Advice**

If symptoms persist, call a physician. Show this safety data sheet to the doctor in attendance.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

|                                           |                                                                                                                          |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Ingestion</b>                          | Do NOT induce vomiting. Get medical attention.                                                                           |
| <b>Inhalation</b>                         | Remove to fresh air. Get medical attention immediately if symptoms occur. If not breathing, give artificial respiration. |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                         |

## **4.2. Most important symptoms and effects, both acute and delayed**

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

## **4.3. Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## **SECTION 5: FIREFIGHTING MEASURES**

### **5.1. Extinguishing media**

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### **5.2. Special hazards arising from the substance or mixture**

Combustible material. Containers may explode when heated. Thermal decomposition can lead to release of irritating gases and vapors.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Sulfides, Formaldehyde.

### **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges. Ensure adequate ventilation.

### **6.2. Environmental precautions**

Should not be released into the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Keep away from open flames, hot surfaces and sources of ignition. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame.

Technical Rules for Hazardous Substances (TRGS) 510 Storage Class (LGK)  
(Germany)

Storage Class/LGK 10

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s):

| Component          | Italy | Germany                                                                                                                                                                                                                                                              | Portugal | The Netherlands | Finland                    |
|--------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|----------------------------|
| Dimethyl sulfoxide |       | TWA: 50 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 160 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 ppm (8 Stunden). MAK<br>TWA: 160 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 100 ppm<br>Höhepunkt: 320 mg/m <sup>3</sup><br>Haut |          |                 | TWA: 50 ppm 8 tunteina lho |

| Component          | Austria                                                                       | Denmark                                                   | Switzerland                                                                                                                                      | Poland | Norway |
|--------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|
| Dimethyl sulfoxide | Haut<br>MAK-TMW: 50 ppm 8 Stunden<br>MAK-TMW: 160 mg/m <sup>3</sup> 8 Stunden | TWA: 50 ppm 8 timer<br>TWA: 160 mg/m <sup>3</sup> 8 timer | Haut/Peau<br>STEL: 100 ppm 15 Minuten<br>STEL: 320 mg/m <sup>3</sup> 15 Minuten<br>TWA: 50 ppm 8 Stunden<br>TWA: 160 mg/m <sup>3</sup> 8 Stunden |        |        |

| Component          | Estonia               | Gibraltar | Greece | Hungary | Iceland |
|--------------------|-----------------------|-----------|--------|---------|---------|
| Dimethyl sulfoxide | Nahk<br>TWA: 50 ppm 8 |           |        |         |         |

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

|  |                                                                                                                                          |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|
|  | tundides.<br>TWA: 150 mg/m <sup>3</sup> 8<br>tundides.<br>STEL: 150 ppm 15<br>minutites.<br>STEL: 500 mg/m <sup>3</sup> 15<br>minutites. |  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|--|

| Component          | Latvia | Lithuania                                                                                                  | Luxembourg | Malta | Romania |
|--------------------|--------|------------------------------------------------------------------------------------------------------------|------------|-------|---------|
| Dimethyl sulfoxide |        | TWA: 50 ppm IPRD<br>TWA: 150 mg/m <sup>3</sup> IPRD<br>Oda<br>STEL: 150 ppm<br>STEL: 500 mg/m <sup>3</sup> |            |       |         |

| Component          | Russia                    | Slovak Republic | Slovenia                                                                                                                                    | Sweden                                                                                                                                                                               | Turkey |
|--------------------|---------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| Dimethyl sulfoxide | MAC: 20 mg/m <sup>3</sup> |                 | TWA: 160 mg/m <sup>3</sup> 8 urah<br>TWA: 50 ppm 8 urah<br>Koža<br>STEL: 100 ppm 15<br>minutah<br>STEL: 320 mg/m <sup>3</sup> 15<br>minutah | Indicative STEL: 150<br>ppm 15 minuter<br>Indicative STEL: 500<br>mg/m <sup>3</sup> 15 minuter<br>TLV: 50 ppm 8 timmar.<br>NGV<br>TLV: 150 mg/m <sup>3</sup> 8<br>timmar. NGV<br>Hud |        |

## Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

## Monitoring methods

BS EN 14042:2003 Title Identifier: Workplace atmospheres. Guide for the application and use of procedures for the assessment of exposure to chemical and biological agents.

MDHS70 General methods for sampling airborne gases and vapours

**Derived No Effect Level (DNEL)** No information available

| <u>Route of exposure</u>     | Acute effects (local) | Acute effects (systemic) | Chronic effects (local) | Chronic effects (systemic)              |
|------------------------------|-----------------------|--------------------------|-------------------------|-----------------------------------------|
| Oral<br>Dermal<br>Inhalation | 265 mg/m <sup>3</sup> |                          | 265 mg/m <sup>3</sup>   | 200 mg/kg bw/d<br>484 mg/m <sup>3</sup> |

**Predicted No Effect Concentration (PNEC)** No information available.

|                      |            |
|----------------------|------------|
| Fresh water          | 17 mg/l    |
| Fresh water sediment | 13.4 mg/kg |
| Marine water         | 1.7 mg/l   |
| Soil (Agriculture)   | 3.02 mg/kg |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

### Personal protective equipment

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

**Eye Protection** Wear safety glasses with side shields (or goggles) (European standard - EN 166)

**Hand Protection** Protective gloves

| Glove material | Breakthrough time | Glove thickness | EU standard       | Glove comments                                                                 |
|----------------|-------------------|-----------------|-------------------|--------------------------------------------------------------------------------|
| Neoprene       | > 480 minutes     | 0.45 mm         | Level 6<br>EN 374 | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Nitrile rubber | > 480 minutes     | > 0.2 mm        |                   |                                                                                |

**Skin and body protection** Long sleeved clothing

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

**Large scale/emergency use** Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Particle filter

**Small scale/Laboratory use** Maintain adequate ventilation

**Environmental exposure controls** Prevent product from entering drains.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                               |                                          |
|------------------------------------------------|-----------------------------------------------|------------------------------------------|
| <b>Physical State</b>                          | Liquid                                        |                                          |
| <b>Appearance</b>                              | Colorless                                     |                                          |
| <b>Odor</b>                                    | Odorless                                      |                                          |
| <b>Odor Threshold</b>                          | No data available                             |                                          |
| <b>Melting Point/Range</b>                     | 18.4 °C / 65.1 °F                             |                                          |
| <b>Softening Point</b>                         | No data available                             |                                          |
| <b>Boiling Point/Range</b>                     | 189 °C / 372.2 °F                             |                                          |
| <b>Flammability (liquid)</b>                   | Combustible liquid                            | On basis of test data                    |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                   |
| <b>Explosion Limits</b>                        | <b>Lower</b> 2.6 Vol%<br><b>Upper</b> 42 Vol% |                                          |
| <b>Flash Point</b>                             | 87 °C / 188.6 °F                              | <b>Method -</b> No information available |
| <b>Autoignition Temperature</b>                | 301 °C / 573.8 °F                             |                                          |
| <b>Decomposition Temperature</b>               | > 190°C                                       |                                          |
| <b>pH</b>                                      | No information available                      |                                          |
| <b>Viscosity</b>                               | 1.98 mPa.s @ 25°C                             |                                          |
| <b>Water Solubility</b>                        | Soluble                                       |                                          |
| <b>Solubility in other solvents</b>            | No information available                      |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                          |
| <b>Component</b>                               | <b>log Pow</b>                                |                                          |
| Dimethyl sulfoxide                             | -2.03                                         |                                          |
| <b>Vapor Pressure</b>                          | 0.55 mbar @ 20°C                              |                                          |
| <b>Density / Specific Gravity</b>              | 1.100                                         |                                          |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                   |
| <b>Vapor Density</b>                           | 2.7                                           | (Air = 1.0)                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                       |                                          |

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

## 9.2. Other information

Molecular Formula C<sub>2</sub> H<sub>6</sub> O S  
Molecular Weight 78.13  
Explosive Properties explosive air/vapour mixtures possible  
Evaporation Rate No information available

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Hygroscopic.

### 10.3. Possibility of hazardous reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions Thermal decomposition can take place above 189°C / 372°F.

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Exposure to moist air or water. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Strong bases. Alkali metals.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Sulfides. Formaldehyde.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component          | LD50 Oral                  | LD50 Dermal | LC50 Inhalation              |
|--------------------|----------------------------|-------------|------------------------------|
| Dimethyl sulfoxide | LD50 = 28300 mg/kg ( Rat ) | -           | LC50 > 5.33 mg/L ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

#### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

(f) carcinogenicity; Based on available data, the classification criteria are not met  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Based on available data, the classification criteria are not met

Target Organs None known.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

#### Ecotoxicity effects

Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Do not empty into drains. .

| Component          | Freshwater Fish                         | Water Flea         | Freshwater Algae            |
|--------------------|-----------------------------------------|--------------------|-----------------------------|
| Dimethyl sulfoxide | 40 g/L LC50 96 h<br>33-37 g/L LC50 96 h | EC50 24h 7000 mg/L | EC50 96h 12350 - 25500 mg/L |

| Component          | Microtox                                                                                                                                                       | M-Factor |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| Dimethyl sulfoxide | = 16000 mg/L EC50 <i>Pseudomonas putida</i> 16 h<br>= 32 g/L EC50 <i>Tetrahymena pyriformis</i> 24 h<br>= 77 mg/L EC50 <i>Photobacterium phosphoreum</i> 5 min |          |

### 12.2. Persistence and degradability

#### Persistence

Persistence is unlikely.

#### Degradation in sewage treatment plant

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

### 12.3. Bioaccumulative potential

Bioaccumulation is unlikely

| Component          | log Pow | Bioconcentration factor (BCF) |
|--------------------|---------|-------------------------------|
| Dimethyl sulfoxide | -2.03   | No data available             |

### 12.4. Mobility in soil

The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

## 12.5. Results of PBT and vPvB assessment

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## 12.6. Endocrine disrupting properties

### Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

#### Waste from Residues/Unused Products

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

#### Contaminated Packaging

Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

#### European Waste Catalogue (EWC)

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### Other Information

Do not flush to sewer.

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### ADR

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

### IATA

Not regulated

#### 14.1. UN number

#### 14.2. UN proper shipping name

#### 14.3. Transport hazard class(es)

#### 14.4. Packing group

#### 14.5. Environmental hazards

No hazards identified

#### 14.6. Special precautions for user

No special precautions required

#### 14.7. Maritime transport in bulk according to IMO instruments

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

## 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed, Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), China (IECSC), Japan (ENCS), Australia (AICS), Korea (ECL).

| Component          | EINECS    | ELINCS | NLP | TSCA | DSL | NDSL | PICCS | ENCS | IECSC | AICS | KECL         |
|--------------------|-----------|--------|-----|------|-----|------|-------|------|-------|------|--------------|
| Dimethyl sulfoxide | 200-664-3 | -      |     | X    | X   | -    | X     | X    | X     | X    | KE-3236<br>7 |

### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

### National Regulations

#### WGK Classification

See table for values

| Component          | Germany - Water Classification (VwVwS) | Germany - TA-Luft Class |
|--------------------|----------------------------------------|-------------------------|
| Dimethyl sulfoxide | WGK1                                   |                         |

| Component          | France - INRS (Tables of occupational diseases)      |
|--------------------|------------------------------------------------------|
| Dimethyl sulfoxide | Tableaux des maladies professionnelles (TMP) - RG 84 |

Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

#### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

# SAFETY DATA SHEET

DIMETHYL SULFOXIDE

Revision Date 03-Jan-2021

BCF - Bioconcentration factor

VOC (volatile organic compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

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Revision Date 03-Jan-2021

Revision Summary Update to CLP Format.

**This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006  
COMMISSION REGULATION (EU) 2020/878 amending Annex II to Regulation (EC) No  
1907/2006**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**